# 5 The $LS$-coupling scheme



In this chapter we shall look at atoms with two valence electrons, e.g. alkaline earth metals such as Mg and Ca. The structures of these elements have many similarities with helium, and we shall also use the central-field approximation that was introduced for the alkalis in the previous chapter. We start with the Hamiltonian for $N$ electrons in eqn 4.2 and insert the expression for the central potential $V_{\mathrm{CF}}(r)$ (eqn 4.3) to give

$$H = \sum_{i=1}^{N}\left[-\frac{\hbar^2}{2m}\nabla_i^2 + V_{\mathrm{CF}}(r_i) + \left\{\sum_{j>i}^{N}\frac{e^2/4\pi\epsilon_0}{r_{ij}} - S(r_i)\right\}\right].$$

This Hamiltonian can be written as $H = H_{\mathrm{CF}} + H_{\mathrm{re}}$, where the central-field Hamiltonian $H_{\mathrm{CF}}$ is that defined in eqn 4.4 and

$$H_{\mathrm{re}} = \sum_{i=1}^{N}\left\{\sum_{j>i}^{N}\frac{e^2/4\pi\epsilon_0}{r_{ij}} - S(r_i)\right\} \tag{5.1}$$

is the residual electrostatic interaction. This represents that part of the repulsion not taken into account by the central field. One might think that the field left over is somehow non-central. This is not necessarily true. For configurations such as 1s2s in He, or 3s4s in Mg, both electrons have spherically-symmetric distributions but a central field cannot completely account for the repulsion between them—a potential $V_{\mathrm{CF}}(r)$ does not include the effect of the correlation of the electrons' positions that leads to the exchange integral.[1] The residual electrostatic interaction perturbs the electronic configurations $n_1l_1n_2l_2$ that are the eigenstates of the central field. These angular momentum eigenstates for the two electrons are products of their orbital and spin functions $|l_1m_{l_1}s_1m_{s_1}\rangle\,|l_2m_{l_2}s_2m_{s_2}\rangle$ and their energy does not depend on the atom's orientation so that all the different $m_l$ states are degenerate, e.g. the configuration 3p4p has $(2l_1+1)(2l_2+1) = 9$ degenerate combinations of $Y_{l_1,m_1}Y_{l_2,m_2}$.[2] Each of these spatial states has four spin functions associated with it, but we do not need to consider thirty-six degenerate states since the problem separates into spatial and spin parts, as in helium. Nevertheless, the direct approach would require diagonalising matrices of larger dimensions than the simple $2\times 2$ matrix whose determinant was given in eqn 3.17. Therefore, instead of that brute-force approach, we use the 'look-before-you-leap' method that starts by finding the eigenstates of the perturbation $H_{\mathrm{re}}$. In that representation, $H_{\mathrm{re}}$ is a diagonal matrix with the eigenvalues as its diagonal elements.

[1] Choosing $S(r)$ to account for all the repulsion between the spherically-symmetric core and the electrons outside the closed shells, and also within the core, leaves the repulsion between the two valence electrons, i.e. $H_{\mathrm{re}} \simeq e^2/4\pi\epsilon_0 r_{12}$. This approximation highlights the similarity with helium (although the expectation value is evaluated with different wavefunctions). Although it simplifies the equations nicely, this is not the best approximation for accurate calculations—$S(r)$ can be chosen to include most of the direct integral (cf. Section 3.3.2). For alkali metal atoms, which we studied in the last chapter, the repulsion between electrons gives a spherically-symmetric potential, so that $H_{\mathrm{re}} = 0$.

[2] For two p-electrons we cannot ignore $m_l$ as we did in the treatment of 1s$nl$ configurations in helium. Configurations with one, or more, s-electrons can be treated in the way already described for helium but with the radial wavefunctions calculated numerically.

The interaction between the electrons, from their electrostatic repulsion, causes their orbital angular momenta to change, i.e. in the vector model $\mathbf{l}_1$ and $\mathbf{l}_2$ change direction, but their magnitudes remain constant. This internal interaction does not change the total orbital angular momentum $\mathbf{L} = \mathbf{l}_1 + \mathbf{l}_2$, so $\mathbf{l}_1$ and $\mathbf{l}_2$ move (or precess) around this vector, as illustrated in Fig. 5.1. When no external torque acts on the atom, $\mathbf{L}$ has a fixed orientation in space so its $z$-component $M_L$ is also a constant of the motion ($m_{l_1}$ and $m_{l_2}$ are not good quantum numbers). This classical picture of conservation of total angular momentum corresponds to the quantum mechanical result that the operators $L^2$ and $L_z$ both commute with $H_{\rm re}$:[3]

$$\left[L^2, H_{\rm re}\right] = 0 \qquad \text{and} \qquad \left[L_z, H_{\rm re}\right] = 0\,. \tag{5.2}$$

Since $H_{\rm re}$ does not depend on spin it must also be true that

$$\left[S^2, H_{\rm re}\right] = 0 \qquad \text{and} \qquad \left[S_z, H_{\rm re}\right] = 0\,. \tag{5.3}$$

Actually, $H_{\rm re}$ also commutes with the individual spins $\mathbf{s}_1$ and $\mathbf{s}_2$ but we chose eigenfunctions of $\mathbf{S}$ to antisymmetrise the wavefunctions, as in helium—the spin eigenstates for two electrons are $\psi^{\rm A}_{\rm spin}$ and $\psi^{\rm S}_{\rm spin}$ for $S = 0$ and 1, respectively.[4] The quantum numbers $L, M_L, S$ and $M_S$ have well-defined values in this **Russell–Saunders** or $LS$-**coupling scheme**. Thus the eigenstates of $H_{\rm re}$ are $|LM_LSM_S\rangle$. In the $LS$-coupling scheme the energy levels labelled by $L$ and $S$ are called **terms** (and there is degeneracy with respect to $M_L$ and $M_S$). We saw examples of $^1L$ and $^3L$ terms for the $1snl$ configurations in helium where the $LS$-coupling scheme is a very good approximation. A more complex example is an $np n'$p configuration, e.g. 3p4p in silicon, that has six terms as follows:

$$\begin{aligned} l_1 &= 1, \quad l_2 = 1 \quad &&\Rightarrow \quad L = 0,\,1 \text{ or } 2\,,\\ s_1 &= \frac{1}{2}, \quad s_2 = \frac{1}{2} \quad &&\Rightarrow \quad S = 0 \text{ or } 1\,;\\ \text{terms:} &\quad {}^{2S+1}L = {}^1\text{S}, {}^1\text{P}, {}^1\text{D}, {}^3\text{S}, {}^3\text{P}, {}^3\text{D}\,. \end{aligned}$$

The direct and exchange integrals that determine the energies of these terms are complicated to evaluate (see Woodgate (1980) for details) and here we shall simply make some empirical observations based on the terms diagrams in Figs 5.2 and 5.3. The $(2l_1+1)(2l_2+1) = 9$ degenerate states of orbital angular momentum become the $1+3+5=9$ states of $M_L$ associated with the S, P and D terms, respectively. As in helium, linear combinations of the four degenerate spin states lead to triplet and one singlet terms but, unlike helium, triplets do not necessarily lie below singlets. Also, the $3\text{p}^2$ configuration has fewer terms than the 3p4p configuration for equivalent electrons, because of the Pauli exclusion principle (see Exercise 5.6).

In the special case of *ground* configurations of *equivalent electrons* the spin and orbital angular momentum of the lowest-energy term follow some empirical rules, called **Hund's rules**: the lowest-energy term has the largest value of $S$ consistent with the Pauli exclusion principle.[5] If

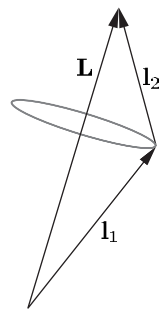



**Fig. 5.1** The residual electrostatic interaction causes $\mathbf{l}_1$ and $\mathbf{l}_2$ to precess around their resultant $\mathbf{L} = \mathbf{l}_1 + \mathbf{l}_2$.

[3] The proof is straightforward for the quantum operator: $L_z = l_{1z} + l_{2z}$ since $m_{l_1} = q$ always occurs with $m_{l_2} = -q$ in eqn 3.30.

[4] The Hamiltonian $H$ commutes with the exchange (or swap) operator $X_{ij}$ that interchanges the labels of the particles $i \leftrightarrow j$; thus states that are simultaneously eigenfunctions of both operators exist. This is obviously true for the Hamiltonian of the helium atom in eqn 3.1 (which looks the same if $1 \leftrightarrow 2$), but it also holds for eqn 5.1. In general, swapping particles with the same mass and charge does not change the Hamiltonian for the electrostatic interactions of a system.

[5] Two electrons cannot both have the same set of quantum numbers.

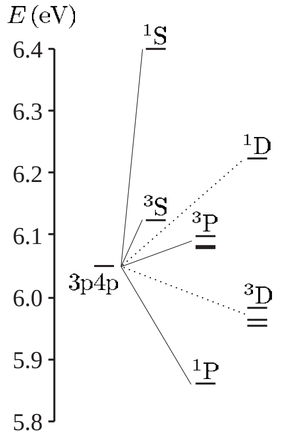



**Fig. 5.2** The terms of the 3p4p configuration in silicon all lie about 6 eV above the ground state. The residual electrostatic interaction leads to energy differences of $\sim 0.2$ eV between the terms, and the fine-structure splitting is an order of magnitude smaller, as indicated for the $^3$P and $^3$D terms. This structure is well described by the *LS*-coupling scheme.

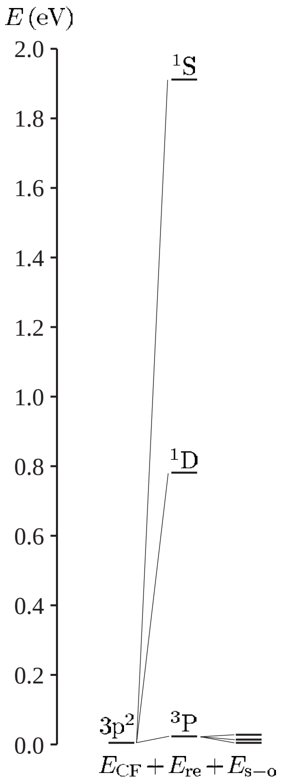



**Fig. 5.3** The energies of terms of the 3p$^2$ configuration of silicon. For equivalent electrons the Pauli exclusion principle restricts the number of terms—there are only three compared to the six in Fig. 5.2. The lowest-energy term is $^3$P, in accordance with Hund's rules, and this is the ground state of silicon atoms.

there are several such terms then the one with the largest $L$ is lowest. The lowest term in Fig. 5.3 is consistent with these rules;[6] the rule says nothing about the ordering of the other terms (or about any of the terms in Fig. 5.2). Configurations of equivalent electrons are especially important since they occur in the ground configuration of elements in the periodic table, e.g. for the 3d$^6$ configuration in iron, Hund's rules give the lowest term as $^5$D (see Exercise 5.6).[7]

[6] Hund's rules are so commonly misapplied that it is worth spelling out that they only apply to the *lowest term* of the *ground configuration* for cases where there is only one incomplete subshell.

[7] The large total spin has important consequences for magnetism (Blundell 2001).

## 5.1 Fine structure in the $LS$-coupling scheme

Fine structure arises from the spin–orbit interaction for each of the unpaired electrons given by the Hamiltonian

$$H_{\mathrm{s-o}} = \beta_1 \mathbf{s}_1 \cdot \mathbf{l}_1 + \beta_2 \mathbf{s}_2 \cdot \mathbf{l}_2 \,.$$

For atoms with two valence electrons $H_{\mathrm{s-o}}$ acts as a perturbation on the states $|LM_L S M_S\rangle$. In the vector model, this interaction between the spin and orbital angular momentum causes $\mathbf{L}$ and $\mathbf{S}$ to change direction, so that neither $L_z$ nor $S_z$ remains constant; but the total electronic angular momentum $\mathbf{J} = \mathbf{L} + \mathbf{S}$, and its $z$-component $J_z$, are both constant because no external torque acts on the atom. We shall now evaluate the effect of the perturbation $H_{\mathrm{s-o}}$ on a term using the vector model. In the vector-model description of the $LS$-coupling scheme, $\mathbf{l}_1$ and $\mathbf{l}_2$ precess around $\mathbf{L}$, as shown in Fig. 5.4; the components perpendicular to this fixed direction average to zero (over time) so that only the component of these vectors along $\mathbf{L}$ needs to be considered, e.g. $\mathbf{l}_1 \to \left\{ \left(\overline{\mathbf{l}_1 \cdot \mathbf{L}}\right) / |\mathbf{L}|^2 \right\} \mathbf{L}$. The time average $\overline{\mathbf{l}_1 \cdot \mathbf{L}}$ in the vector model becomes the expectation value $\langle \mathbf{l}_1 \cdot \mathbf{L} \rangle$ in quantum mechanics; also we have to use $L(L+1)$ for the magnitude-squared of the vector. Applying the same projection procedure to the spins leads to

$$\begin{aligned} H_{\mathrm{s-o}} &= \beta_1 \frac{\langle \mathbf{s}_1 \cdot \mathbf{S} \rangle}{S(S+1)} \mathbf{S} \cdot \frac{\langle \mathbf{l}_1 \cdot \mathbf{L} \rangle}{L(L+1)} \mathbf{L} + \beta_2 \frac{\langle \mathbf{s}_2 \cdot \mathbf{S} \rangle}{S(S+1)} \mathbf{S} \cdot \frac{\langle \mathbf{l}_2 \cdot \mathbf{L} \rangle}{L(L+1)} \mathbf{L} \\ &= \beta_{LS} \, \mathbf{S} \cdot \mathbf{L} \,. \end{aligned} \tag{5.4}$$

The derivation of this equation by the vector model that argues by analogy with classical vectors can be fully justified by reference to the theory of angular momentum. It can be shown that, in the basis $|J\,M_J\rangle$ of the eigenstates of a general angular momentum operator $\mathbf{J}$ and its component $J_z$, the matrix elements of any vector operator $\mathbf{V}$ are proportional to those of $\mathbf{J}$, i.e. $\langle J\,M_J|\,\mathbf{V}\,|J\,M_J\rangle = c\,\langle J\,M_J|\,\mathbf{J}\,|J\,M_J\rangle$.[8] Figure 5.5 gives a pictorial representation of why it is only the component of $\mathbf{V}$ along $\mathbf{J}$ that is well defined. We want to apply this result to the case where $\mathbf{V} = \mathbf{l}_1$ or $\mathbf{l}_2$ in the basis of eigenstates $|L\,M_L\rangle$, and analogously for the spins. For $\langle L\ M_L|\,\mathbf{l}_1\,|L\ M_L\rangle = c\,\langle L\ M_L|\,\mathbf{L}\,|L\ M_L\rangle$ the constant $c$ is determined by taking the dot product of both sides with $\mathbf{L}$ to give

$$c = \frac{\langle L\,M_L|\,\mathbf{l}_1 \cdot \mathbf{L}\,|L\,M_L\rangle}{\langle L\,M_L|\,\mathbf{L} \cdot \mathbf{L}\,|L\,M_L\rangle}\,;$$

hence

$$\langle L\ M_L|\,\mathbf{l}_1\,|L\ M_L\rangle = \frac{\langle \mathbf{l}_1 \cdot \mathbf{L} \rangle}{L(L+1)} \langle L\ M_L|\,\mathbf{L}\,|L\ M_L\rangle \,. \tag{5.5}$$

This is an example of the projection theorem and can also be applied to $\mathbf{l}_2$ and to $\mathbf{s}_1$ and $\mathbf{s}_2$ in the basis of eigenstates $|S\,M_S\rangle$. It is clear that, for diagonal matrix elements, these quantum mechanical results give the same result of the vector model.

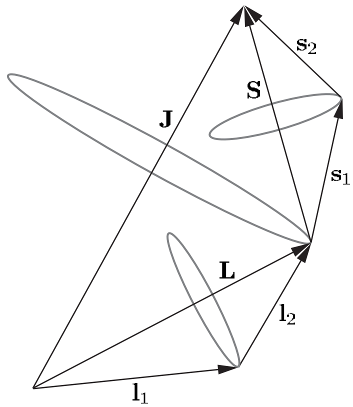



**Fig. 5.4** In the $LS$-coupling scheme the orbital angular momenta of the two electrons couple to give total angular momentum $\mathbf{L} = \mathbf{l}_1 + \mathbf{l}_2$. In the vector model $\mathbf{l}_1$ and $\mathbf{l}_2$ precess around $\mathbf{L}$; similarly, $\mathbf{s}_1$ and $\mathbf{s}_2$ precess around $\mathbf{S}$. $\mathbf{L}$ and $\mathbf{S}$ precess around the total angular momentum $\mathbf{J}$ (but more slowly than the precession of $\mathbf{l}_1$ and $\mathbf{l}_2$ around $\mathbf{L}$ because the spin–orbit interaction is 'weaker' than the residual electrostatic interaction).

[8] This is particular case of a more general result called the Wigner–Eckart theorem which is the cornerstone of the theory of angular momentum. This powerful theorem also applies to off-diagonal elements such as $\langle J\,M_J|\,\mathbf{V}\,|J\,M_J'\rangle$, and to more complicated operators such as those for quadrupole moments. It is used extensively in advanced atomic physics—see the 'Further reading' section in this chapter.

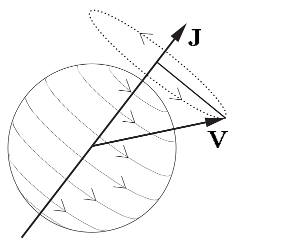



**Fig. 5.5** A pictorial representation of the project theorem for an atom, where $\mathbf{J}$ defines the axis of the system.

Equation 5.4 has the same form as the spin–orbit interaction for the single-electron case but with capital letters rather than $\mathbf{s}\cdot\mathbf{l}$. The constant $\beta_{LS}$ that gives the spin–orbit interaction for each term is related to that for the individual electrons (see Exercise 5.2). The energy shift is

$$E_{\mathrm{s-o}} = \beta_{LS} \langle \mathbf{S} \cdot \mathbf{L} \rangle \,. \tag{5.6}$$

To find this energy we need to evaluate the expectation value of the operator $\mathbf{L} \cdot \mathbf{S} = (\mathbf{J} \cdot \mathbf{J} - \mathbf{L} \cdot \mathbf{L} - \mathbf{S} \cdot \mathbf{S})/2$ for each term $^{2S+1}L$. Each term has $(2S+1)(2L+1)$ degenerate states. Any linear combination of these states is also an eigenstate with the same electrostatic energy and we can use this freedom to choose suitable eigenstates and make the calculation of the (magnetic) spin–orbit interaction straightforward. We shall use the states $|LSJM_J\rangle$; these are linear combinations of the basis states $|LM_LSM_S\rangle$ but we do not need to determine their exact form to find the eigenenergies.[9] Evaluation of eqn 5.6 with the states $|LSJM_J\rangle$ gives

$$E_{\mathrm{s-o}} = \frac{\beta_{LS}}{2} \left\{ J\,(J+1) - L\,(L+1) - S\,(S+1) \right\} . \tag{5.7}$$

Thus the energy interval between adjacent $J$ levels is

$$\Delta E_{\mathrm{FS}} = E_J - E_{J-1} = \beta_{LS}\, J \,. \tag{5.8}$$

This is called the interval rule. For example, a $^3$P term ($L = 1 = S$) has three $J$ levels: $^{2S+1}L_J = {}^3\mathrm{P}_0$, $^3\mathrm{P}_1$, $^3\mathrm{P}_2$ (see Fig. 5.6); and the separation between $J = 2$ and $J = 1$ is twice that between $J = 1$ and $J = 0$. The existence of an interval rule in the fine structure of a two-electron system generally indicates that the $LS$-coupling scheme is a good approximation (see the 'Exercises' in this chapter); however, the converse is not true. The $LS$-coupling scheme gives a very accurate description of the energy levels of helium but the fine structure does not exhibit an interval rule (see Example 5.2 later in this chapter).

It is important not to confuse $LS$-coupling (or Russell–Saunders coupling) with the interaction between $\mathbf{L}$ and $\mathbf{S}$ given by $\beta_{LS}\,\mathbf{S}\cdot\mathbf{L}$. In this book the word *interaction* is used for real physical effects described by a Hamiltonian and *coupling* refers to the forming of linear-combination wavefunctions that are eigenstates of angular momentum operators, e.g. eigenstates of $\mathbf{L}$ and $\mathbf{S}$. The $LS$-coupling scheme breaks down as the strength of the interaction $\beta_{LS}\,\mathbf{S}\cdot\mathbf{L}$ increases relative to that of $H_{\mathrm{re}}$.[10]

[9] Similarly, in the one-electron case we found the fine structure without determining the eigenstates $|lsjm_j\rangle$ explicitly in terms of the $Y_{l,m}$ and spin functions.

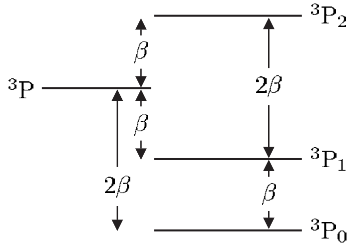



**Fig. 5.6** The fine structure of a $^3$P term obeys the interval rule.

[10] In classical mechanics the word 'coupling' is commonly used more loosely, e.g. for coupled pendulums, or coupled oscillators, the 'coupling between them' is taken to mean the 'interaction between them' that leads to their motions being coupled. (This coupling may take the form of a physical linkage such as a rod or spring between the two systems.)

## 5.2 The $jj$-coupling scheme

To calculate the fine structure in the $LS$-coupling scheme we treated the spin–orbit interaction as a perturbation on a term, $^{2S+1}L$. This is valid when $E_{\mathrm{re}} \gg E_{\mathrm{s-o}}$, which is generally true in light atoms.[11] The spin–orbit interaction increases with atomic number (eqn 4.13) so that it can be similar to $E_{\mathrm{re}}$ for heavy atoms—see Fig. 5.7. However, it is only in cases with particularly small exchange integrals that $E_{\mathrm{s-o}}$ exceeds $E_{\mathrm{re}}$, so that the spin–orbit interaction must be considered before the residual

[11] $E_{\mathrm{s-o}} \sim \beta_{LS}$ and $E_{\mathrm{re}}$ is comparable to the exchange integral.

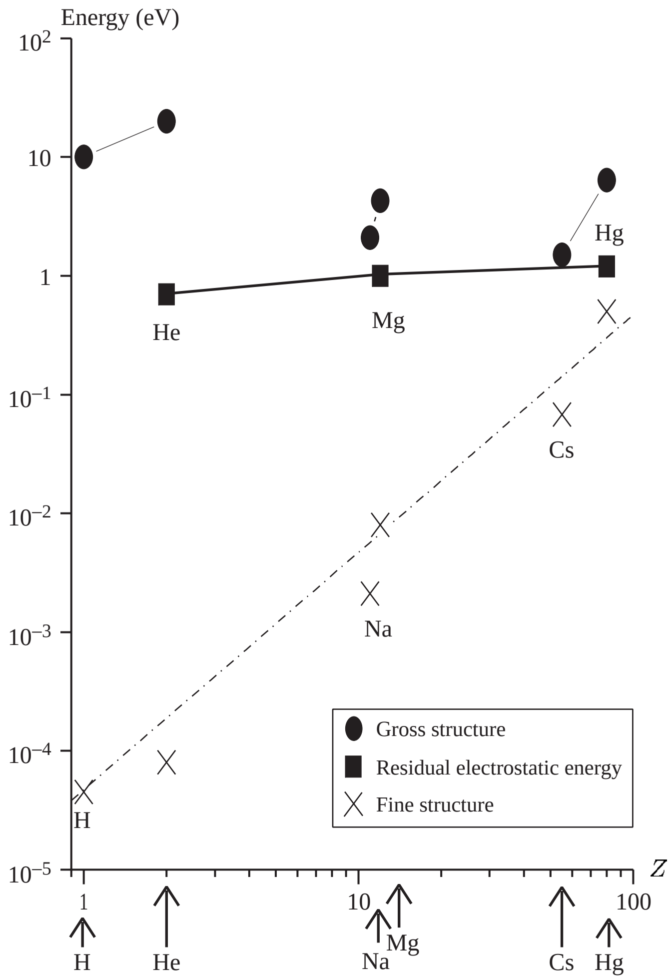



**Fig. 5.7** A plot of typical energies as a function of the atomic number $Z$ (on logarithmic scales). A characteristic energy for the gross structure is taken as the energy required to excite an electron from the ground state to the first excited state—this is less than the ionization energy but has a similar variation with $Z$. The residual electrostatic interaction is the singlet–triplet separation of the lowest excited configuration in some atoms with two valence electrons. The fine structure is the splitting of the lowest p configuration. For all cases the plotted energies are fairly close to the maximum for that type of structure in neutral atoms—higher-lying configurations have smaller values.

electrostatic interaction. When $H_{\mathrm{s-o}}$ acts directly on a configuration it causes the $\mathbf{l}$ and $\mathbf{s}$ of each individual electron to be coupled together to give $\mathbf{j}_1 = \mathbf{l}_1 + \mathbf{s}_1$ and $\mathbf{j}_2 = \mathbf{l}_2 + \mathbf{s}_2$; in the vector model this corresponds to $\mathbf{l}$ and $\mathbf{s}$ precessing around $\mathbf{j}$ independently of the other electrons. In this $jj$-coupling scheme each valence electron acts on its own, as in alkali atoms. For an sp configuration the s-electron can only have $j_1 = 1/2$ and the p-electron has $j_2 = 1/2$ or $3/2$; so there are two levels, denoted by $(j_1,\ j_2) = (1/2, 1/2)$ and $(1/2, 3/2)$. The residual electrostatic interaction acts as a perturbation on the $jj$-coupled levels; it causes the angular momenta of the electrons to be coupled to give total angular momentum $\mathbf{J} = \mathbf{j}_1 + \mathbf{j}_2$ (as illustrated in Fig. 5.8). Since there is no external torque on the atom, $M_J$ is also a good quantum number. For an sp configuration there are pairs of $J$ levels for each of the two original $jj$-coupled levels, e.g. $(j_1,\ j_2)_J = (1/2, 1/2)_0\,,\ (1/2, 1/2)_1$ and $(1/2, 3/2)_1\,, (1/2, 3/2)_2$. This doublet structure, shown in Fig. 5.10, contrasts with the singlets and triplets in the $LS$-coupling scheme.

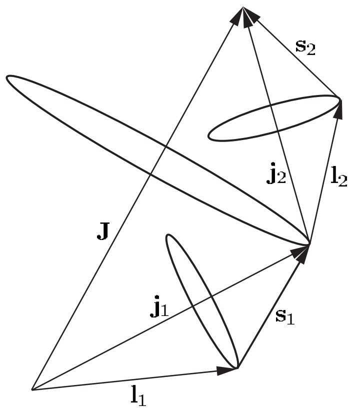



**Fig. 5.8** The $jj$-coupling scheme. The spin–orbit interaction energy is large compared to the $E_{\mathrm{re}}$. (Cf. Fig. 5.4 for the $LS$-coupling scheme.)

[12]In both of these cases we assume an isolated configuration, i.e. that the energy separation of the different configurations in the central field is greater than the perturbation produced by either $E_{\mathrm{s-o}}$ or $E_{\mathrm{re}}$.

In summary, the conditions for $LS$- and $jj$-coupling are as follows:[12]

$$\begin{aligned} LS\text{-coupling scheme:} \quad & E_{\mathrm{re}} \gg E_{\mathrm{s-o}}\,, \\ jj\text{-coupling scheme:} \quad & E_{\mathrm{s-o}} \gg E_{\mathrm{re}}\,. \end{aligned}$$

## 5.3 Intermediate coupling: the transition between coupling schemes

In this section we shall look at examples of angular momentum coupling schemes in two-electron systems. Figure 5.9 shows energy-level diagrams of Mg and Hg and the following example looks at the structure of these atoms.

**Example 5.1**

| 3s3p, Mg | 6s6p, Hg |
|---|---|
| 2.1850 | 3.76 |
| 2.1870 | 3.94 |
| 2.1911 | 4.40 |
| 3.5051 | 5.40 |

The table gives the energy levels, in units of $10^6\,\mathrm{m}^{-1}$ measured from the ground state, for the 3s3p configuration in magnesium ($Z = 12$) and 6s6p in mercury ($Z = 80$). We shall use these data to identify the levels and assign further quantum numbers.

For an sp configuration we expect $^1$P and $^3$P terms. In the case of magnesium we see that the spacings between the three lowest levels are $2000\,\mathrm{m}^{-1}$ and $4100\,\mathrm{m}^{-1}$; these are close to the 1 to 2 ratio expected from the interval rule for the levels with $J = 0, 1$ and 2 that arise from the triplet. The $LS$-coupling scheme gives an accurate description because

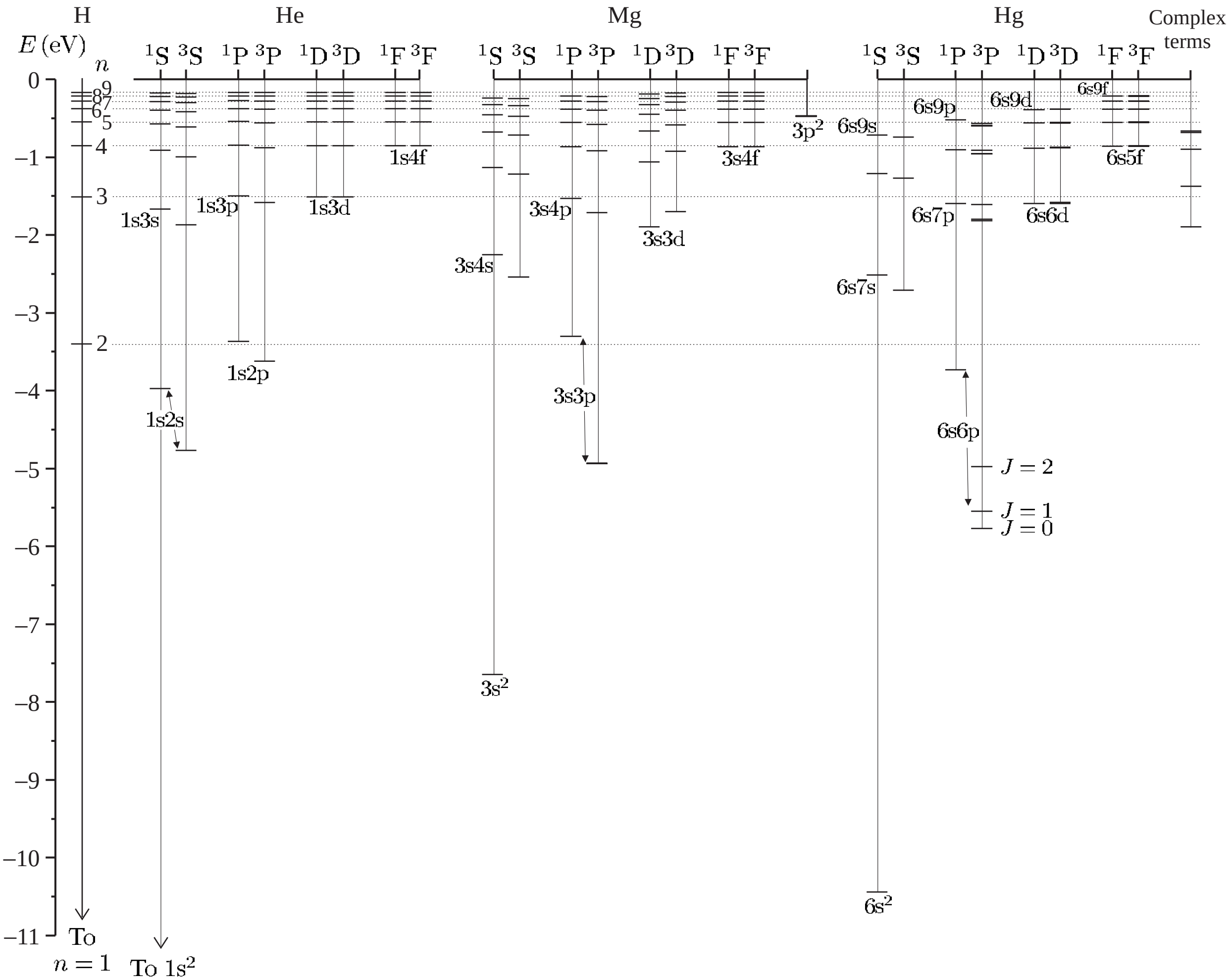



**Fig. 5.9** The terms of helium, magnesium and mercury are plotted on the same energy scale (with hydrogen on the left for comparison). The fine structure of the lighter atoms is too small to be seen on this scale and the $LS$-coupling scheme gives a very accurate description. This scheme gives an approximate description for the low-lying terms of mercury even though it has a much larger fine structure, e.g. for the 6s6p configuration the $E_{\text{re}} > E_{\text{s-o}}$ but the interval rule is not obeyed because the spin–orbit interaction is not very small compared to the residual electrostatic interaction. The $1s^2$ configuration of helium is not shown; it has a binding energy of $-24.6\,\text{eV}$ (see Fig. 3.4). The 1s2s and 1s2p configurations of helium lie close to the $n = 2$ shell in hydrogen, and similarly the $1s3l$ configurations lie close to the $n = 3$ shell. In magnesium, the terms of the $3snf$ configurations have very similar energies to those in hydrogen, but the differences get larger as $l$ decreases. The energies of the terms in mercury have large differences from the hydrogen energy levels. Much can be learnt by carefully studying this term diagram, e.g. there is a $^1$P term which has similar energy in the three configurations: 1s2p, 3s3p and 6s6p in He, Mg and Hg, respectively—thus the effective quantum number $n^*$ is similar despite the increase in $n$. Complex terms arise when both valence electrons are excited in Mg, e.g. the $3p^2$ configuration, and the $5d^9 6s^2 6p$ configuration in Hg.

the fine structure is much smaller than the energy separation ($E_{\rm re} \sim 1.3 \times 10^6\,\rm m^{-1}$) between the $^3$P term at $\sim 2.2 \times 10^6\,\rm m^{-1}$ and the $^1\rm P_1$ level at $3.5 \times 10^6\,\rm m^{-1}$. In mercury the spacings of the levels, going down the table, are 0.18, 0.46 and 1.0 (in units of $10^6$ m$^{-1}$); these levels are not so clearly separated into a singlet and triplet. Taking the lowest three levels as $^3\rm P_0$, $^3\rm P_1$ and $^3\rm P_2$ we see that the interval rule is not well obeyed since $0.46/0.18 = 2.6$ (not 2).[13] This deviation from the $LS$-coupling scheme is hardly surprising since this configuration has a spin–orbit interaction only slightly smaller than the singlet–triplet separation. However, even for this heavy atom, the $LS$-coupling scheme gives a closer approximation than the $jj$-coupling scheme.

[13] This identification of the levels is supported by other information, e.g. determination of $J$ from the Zeeman effect and the theoretically predicted behaviour of an sp configuration shown in Fig. 5.10.

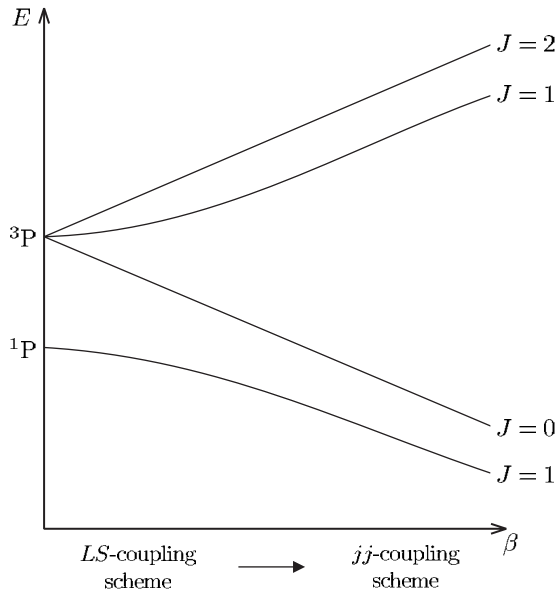



**Fig. 5.10** A theoretical plot of the energy levels that arise from an sp configuration as a function of the strength of the spin–orbit interaction parameter $\beta$ (of the p-electron defined in eqn 2.55). For $\beta = 0$ the two terms, $^3$P and $^1$P, have an energy separation equal to twice the exchange integral; this residual electrostatic energy is assumed to be constant and only $\beta$ varies in the plot. As $\beta$ increases the fine structure of the triplet becomes observable. As $\beta$ increases further the spin–orbit and residual electrostatic interactions become comparable and the $LS$-coupling scheme ceases to be a good approximation: the interval rule and ($LS$-coupling) selection rules break down (as in mercury, see Fig. 5.9). At large $\beta$ the $jj$-coupling scheme is appropriate. The operator $\mathbf{J}$ commutes with $H_{\rm s-o}$ (and $H_{\rm re}$); therefore $H_{\rm s-o}$ only mixes levels of the same $J$, e.g. the two $J = 1$ levels in this case. (The energies of the $J = 0$ and 2 levels are straight lines because their wavefunctions do not change.) Exercise 5.8 gives an example of this transition between the two coupling schemes for $n\rm p(n+1)s$ configurations with $n = 3$ to 5 (that have small exchange integrals).

**Example 5.2** *The* 1s2p *configuration in helium*

| $J$ | $E$ (m$^{-1}$) |
|---|---|
| 2 | 16 908 687 |
| 1 | 16 908 694 |
| 0 | 16 908 793 |
| 1 | 17 113 500 |

The table gives the values of $J$ and the energy, in units of m$^{-1}$ measured from the ground state, for the levels of the 1s2p configuration in helium. The $^3$P term has a fine-structure splitting of about 100 m$^{-1}$ that is much smaller than the singlet–triplet separation of $10^6$ m$^{-1}$ from the electrostatic interaction (twice the exchange integral). Thus the $LS$-coupling scheme gives an excellent description of the helium atom and the selection rules in Table 5.1 are well obeyed. But the interval rule is not obeyed—the intervals between the $J$ levels are 7 m$^{-1}$ and 99 m$^{-1}$ and the fine structure is inverted. This occurs in helium because spin–spin and spin–other-orbit interactions have an energy comparable with that of the spin–orbit interaction.[14] However, for atoms other than helium, the rapid increase in the strength of the spin–orbit interaction with $Z$ ensures that $H_{\mathrm{s-o}}$ dominates over the others. Therefore the fine structure of atoms in the $LS$-coupling scheme usually leads to an interval rule.

[14]The spin–spin interaction arises from the interaction between two magnetic dipoles (independent of any relative motion). See eqn 6.12 and its explanation.

Further examples of energy levels are given in the exercises at the end of this chapter. Figure 5.10 shows a theoretical plot of the transition from the $LS$- to the $jj$-coupling scheme for an sp configuration. Conservation of the total angular momentum means that $J$ is a good quantum number even in the intermediate coupling regime and can always be used to label the levels. The notation $^{2S+1}L_J$ for the $LS$-coupling scheme is often used even for systems in the intermediate regime and also for one-electron systems, e.g. 1s $^2\mathrm{S}_{1/2}$ for the ground state of hydrogen.

**Table 5.1** Selection rules for electric dipole (E1) transitions in the $LS$-coupling scheme. Rules 1–4 apply to all electric dipole transitions; rules 5 and 6 are obeyed only when $L$ and $S$ are good quantum numbers. The right-hand column gives the structure to which the rule applies.

| | | | |
|---|---|---|---|
| 1 | $\Delta J = 0, \pm 1$ | $(J = 0 \not\leftrightarrow J' = 0)$ | Level |
| 2 | $\Delta M_J = 0, \pm 1$ | $(M_J = 0 \not\leftrightarrow M_{J'} = 0$ if $\Delta J = 0)$ | State |
| 3 | Parity changes | | Configuration |
| 4 | $\Delta l = \pm 1$ | One electron jump | Configuration |
| 5 | $\Delta L = 0, \pm 1$ | $(L = 0 \not\leftrightarrow L' = 0)$ | Term |
| 6 | $\Delta S = 0$ | | Term |

## 5.4 Selection rules in the $LS$-coupling scheme

Table 5.1 gives the selection rules for electric dipole transitions in the $LS$-coupling scheme (listed approximately in the order of their strictness). The rule for $J$ reflects the conservation of this quantity and is strictly obeyed; it incorporates the rule for $\Delta j$ in eqn 2.59, but with the additional restriction $J = 0 \not\leftrightarrow J' = 0$ that affects the levels with $J = 0$ that occur in atoms with more than one valence electron. The rule for $\Delta M_J$ follows from that for $\Delta J$: the emission, or absorption, of a photon cannot change the component along the $z$-axis by more than the change in the total atomic angular momentum. (This rule is relevant when the states are resolved, as in the Zeeman effect described in the following section.)[15] The requirement for an overall change in parity and the selection rule for orbital angular momentum were discussed in Section 2.2. In a configuration $n_1l_1\, n_2l_2\, n_3l_3 \cdots n_xl_x$ only one electron changes its value of $l$ (and may also change $n$). The rule for $\Delta L$ allows transitions such as 3p4s $^3\mathrm{P}_1$–3p4p $^3\mathrm{P}_1$. The selection rule $\Delta S = 0$ arises because the electric dipole operator does not act on spin, as noted in Chapter 3 on helium; as a consequence, singlets and triplets form two unconnected sets of energy levels, as shown in Fig. 3.5. Similarly, the singlet and triplet terms of magnesium shown in Fig. 5.9 could be rearranged. In the mercury atom, however, transitions with $\Delta S = 1$ occur, such as $6\mathrm{s}^2\ ^1\mathrm{S}_0$–6s6p $^3\mathrm{P}_1$, that gives a so-called intercombination line with a wavelength of 254 nm.[16] This arises because this heavy atom is not accurately described by the $LS$-coupling scheme and the spin–orbit interaction mixes some $^1\mathrm{P}_1$ wavefunction into the wavefunction for the term that has been labelled $^3\mathrm{P}_1$ (this being its major component). Although not completely forbidden, the rate of this transition is considerably less than it would be for a fully-allowed transition at the same wavelength; however, the intercombination line from a mercury lamp is strong because many of the atoms excited to triplet terms will decay back to the ground state via this transition (see Fig. 5.9).[17]

[15] There is no simple physical explanation of why an $M_J = 0$ to $M_{J'} = 0$ transition does not occur if $J = J'$; it is related to the symmetry of the dipole matrix element $\langle \gamma J\, M_J = 0|r|\gamma' J\, M_J = 0\rangle$, where $\gamma$ and $\gamma'$ represent the other quantum numbers. The particular case of $J = J' = 1$ and $\Delta M_J = 0$ is discussed in Budker *et al.* (2003).

[16] This line comes from the second level in the table given in Example 5.1, since $0.254\,\mu\mathrm{m} = 1/(3.941 \times 10^6\ \mathrm{m}^{-1})$.

[17] Intercombination lines are not observed in magnesium and helium. The relative strength of the intercombination lines and allowed transitions are tabulated in Kuhn (1969).

## 5.5 The Zeeman effect

The Zeeman effect for atoms with a single valence electron was not presented in earlier chapters to avoid repetition and that case is covered by the general expression derived here for the $LS$-coupling scheme.[18] The atom's magnetic moment has orbital and spin contributions (see Blundell 2001, Chapter 2):

$$\boldsymbol{\mu} = -\mu_{\mathrm{B}}\mathbf{L} - g_s\mu_{\mathrm{B}}\mathbf{S}\,. \tag{5.9}$$

The interaction of the atom with an external magnetic field is described by $H_{\mathrm{ZE}} = -\boldsymbol{\mu}\cdot\mathbf{B}$. The expectation value of this Hamiltonian can be calculated in the basis $|LSJM_J\rangle$, provided that $E_{\mathrm{ZE}} \ll E_{\mathrm{s-o}} \ll E_{\mathrm{re}}$, i.e. the interaction can be treated as a perturbation to the fine-structure

[18] Most quantum mechanics texts describe the anomalous Zeeman effect for a single valence electron that applies to the alkalis and hydrogen.

levels of the terms in the $LS$-coupling scheme. In the vector model we project the magnetic moment onto $\mathbf{J}$ (see Fig. 5.11) following the same rules as are used in treating fine structure in the $LS$-coupling scheme (and taking $\mathbf{B} = B\widehat{\mathbf{e}}_z$). This gives

$$H_{\mathrm{ZE}} = -\frac{\langle \boldsymbol{\mu} \cdot \mathbf{J} \rangle}{J\,(J+1)} \mathbf{J} \cdot \mathbf{B} = \frac{\langle \mathbf{L} \cdot \mathbf{J} \rangle + g_s \langle \mathbf{S} \cdot \mathbf{J} \rangle}{J\,(J+1)} \mu_{\mathrm{B}} B J_z \,. \tag{5.10}$$

In the vector model the quantities in angled brackets are time averages.[19] In a quantum description treatment the quantities $\langle \cdots \rangle$ are expectation values of the form $\langle J\,M_J | \cdots | J\,M_J \rangle$.[20] In the vector model

$$E_{\mathrm{ZE}} = g_J \mu_{\mathrm{B}} B M_J \,, \tag{5.11}$$

where the Landé $g$-factor is $g_J = \{\langle \mathbf{L} \cdot \mathbf{J} \rangle + g_s \langle \mathbf{S} \cdot \mathbf{J} \rangle\} / \{J\,(J+1)\}$ . Assuming that $g_s \simeq 2$ (see Section 2.3.4) gives

$$g_J = \frac{3}{2} + \frac{S\,(S+1) - L\,(L+1)}{2J\,(J+1)} \,. \tag{5.12}$$

Singlet terms have $S = 0$ so $\mathbf{J} = \mathbf{L}$ and $g_J = 1$ (no projection is necessary). Thus singlets all have the same Zeeman splitting between $M_J$ states and transitions between singlet terms exhibit the normal Zeeman effect (shown in Fig. 5.12). The $\Delta M_J = \pm 1$ transitions have frequencies shifted by $\pm \mu_{\mathrm{B}} B / h$ with respect to the $\Delta M_J = 0$ transitions.

In atoms with two valence electrons the transitions between triplet terms exhibit the anomalous Zeeman effect. The observed pattern depends on the values of $g_J$ and $J$ for the upper and lower levels, as shown in Fig. 5.13. In both the normal and anomalous effects the $\pi$-transitions ($\Delta M_J = 0$) and $\sigma$-transitions ($\Delta M_J = \pm 1$) have the same polarizations as in the classical model in Section 1.8. Other examples in Exercises 5.10 to 5.12 show how observation of the Zeeman pattern gives information about the angular momentum coupling in the atom. (The Zeeman effect observed for the ${}^2\mathrm{P}_{1/2}$–${}^2\mathrm{S}_{1/2}$ and ${}^2\mathrm{P}_{3/2}$–${}^2\mathrm{S}_{1/2}$ transitions that arise between the fine-structure components of the alkalis and hydrogen is treated in Exercise 5.13.) Exercise 5.14 goes through the Paschen–Back effect that occurs in a strong external magnetic field—see Fig. 5.14.

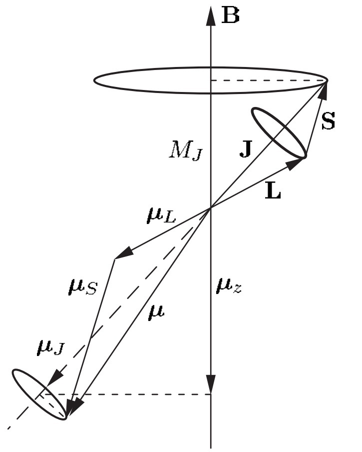



**Fig. 5.11** The projection of the contributions to the total magnetic moment from the orbital motion and the spin are projected along $\mathbf{J}$.

[19] Components perpendicular to $\mathbf{J}$ time-average to zero.

[20] This statement is justified by the projection theorem (Section 5.1), derived from the more general Wigner–Eckart theorem. The theorem shows that the expectation value of the vector operator $\mathbf{L}$ is proportional to that of $\mathbf{J}$ in the basis of eigenstates $|J\,M_J\rangle$, i.e.

$$\langle J\,M_J | \,\mathbf{L}\, | J\,M_J \rangle \propto \langle J\,M_J | \,\mathbf{J}\, | J\,M_J \rangle \,,$$

and similarly for the expectation value of $\mathbf{S}$. The component along the magnetic field is found by taking the dot product with $\mathbf{B}$:

$$\begin{aligned} &\langle J\,M_J | \,\mathbf{L} \cdot \mathbf{B}\, | J\,M_J \rangle \\ &\qquad \propto \langle J\,M_J | \,\mathbf{J} \cdot \mathbf{B}\, | J\,M_J \rangle \\ &\qquad \propto \langle J\,M_J | \,J_z\, | J\,M_J \rangle = M_J \,. \end{aligned}$$

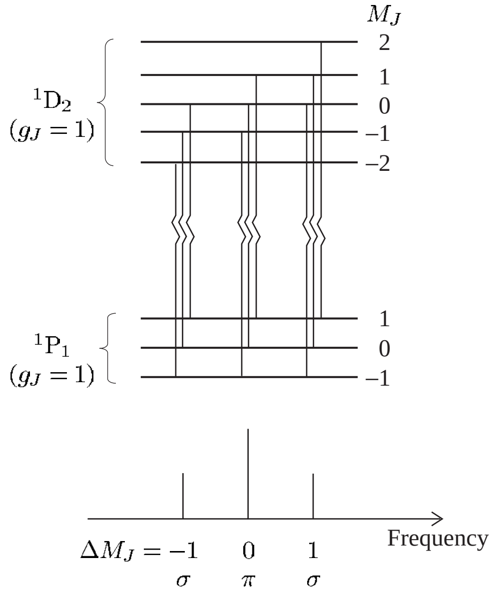



**Fig. 5.12** The normal Zeeman effect on the 1s2p $^1P_1$–1s3d $^1D_2$ line in helium. These levels split into three and five $M_J$ states, respectively. Both levels have $S = 0$ and $g_J = 1$ so that the allowed transitions between the states give the same pattern of three components as the classical model (in Section 1.8)—this is the historical reason why it is called the normal Zeeman effect. Spectroscopists called any other pattern an anomalous Zeeman effect, although such patterns have a straightforward explanation in quantum mechanics and arise whenever $S \neq 0$, e.g. all atoms with one valence electron have $S = 1/2$. The $\pi$- and $\sigma$-components arise from $\Delta M_J = 0$ and $\Delta M_J = \pm 1$ transitions, respectively. (In this example of the normal Zeeman effect each component corresponds to three allowed electric dipole transitions with the *same* frequency but they are drawn with a slight horizontal separation for clarity.)

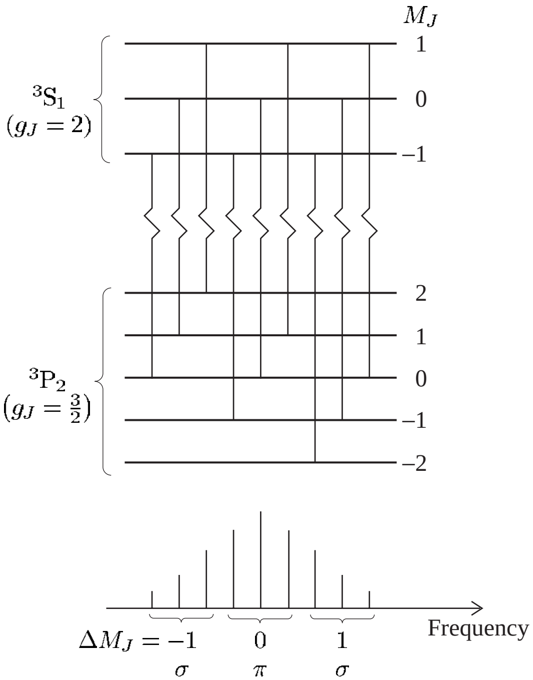



**Fig. 5.13** The anomalous Zeeman effect for the 6s6p $^3P_2$–6s7s $^3S_1$ transition in Hg. The lower and upper levels both have the same number of Zeeman sub-levels (or $M_J$ states) as the levels in Fig. 5.12, but give rise to nine separate components because the levels have different values of $g_J$. (The 6s7s configuration happens to have higher energy than 6s6p, as shown in Fig. 5.9, but the Zeeman pattern does not depend on the relative energy of the levels.)

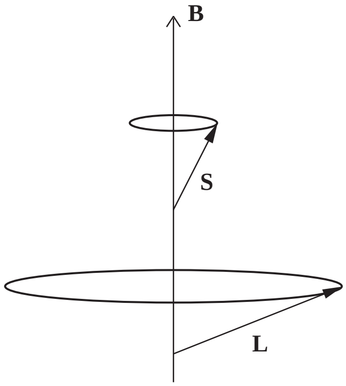



**Fig. 5.14** The Paschen–Back effect occurs in a strong external magnetic field. The spin and orbital angular momentum precess independently about the magnetic field. The states have energies given by $E_{\mathrm{PB}} = \mu_{\mathrm{B}} B (M_L + 2M_S)$.

## 5.6 Summary

Figure 5.15 shows the different layers of structure in a case where the $LS$-coupling scheme is a good approximation, i.e. the residual electrostatic interaction dominates the two magnetic interactions (spin–orbit and with an external magnetic field). The spin–orbit interaction splits terms into different $J$ levels. The Zeeman effect of a weak magnetic field splits the levels into states of given $M_J$, that are also referred to as Zeeman sub-levels.

There are various ways in which this simple picture can break down.

(a) Configuration mixing occurs if the residual electrostatic interaction

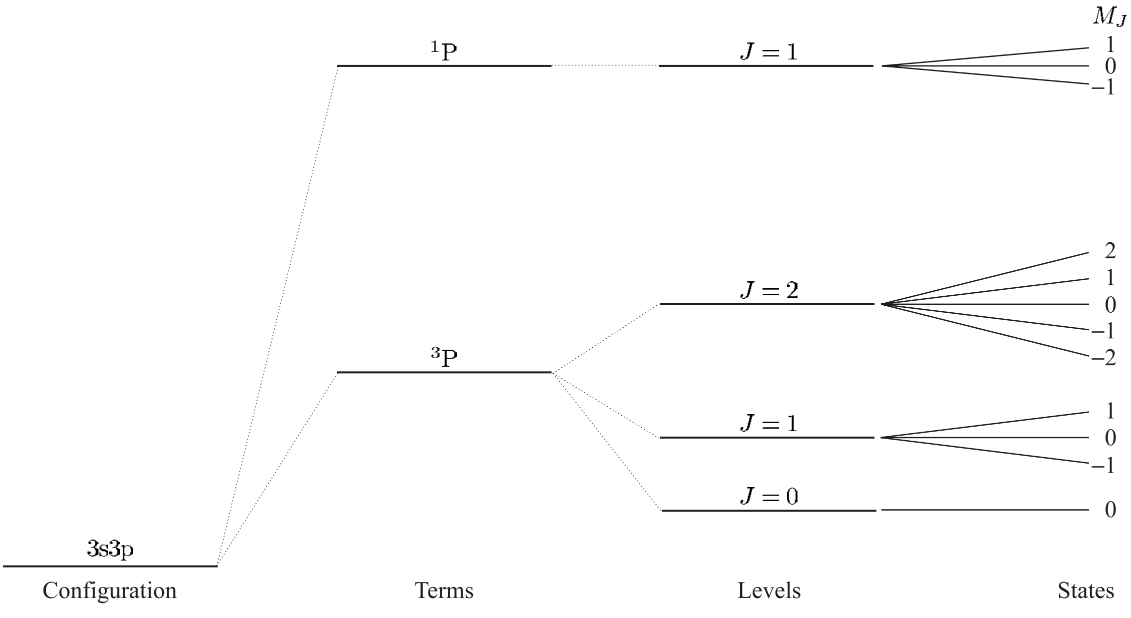



**Fig. 5.15** The hierarchy of atomic structure for the 3s3p configuration of an alkaline earth metal atom.

is not small compared to the energy gap between the configurations—this is common in atoms with complex electronic structure.

(b) The $jj$-coupling scheme is a better approximation than $LS$-coupling or Russell–Saunders coupling when the spin–orbit interaction is greater than the residual electrostatic interaction.

(c) The Paschen–Back effect arises when the interaction with an external magnetic field is stronger than the spin–orbit interaction (with the internal field). This condition is difficult to achieve except for atoms with a low atomic number and hence small fine structure. Similar physics arises in the study of the Zeeman effect on hyperfine structure where the transition between the low-field and high-field regimes occurs at values of the magnetic field that are easily accessible in experiments (see Section 6.3).

## Further reading

The mathematical methods that describe the way in which angular momenta couple together form the backbone of the theory of atomic structure. In this chapter the quantum mechanical operators have been treated by analogy with classical vectors (the vector model) and the Wigner–Eckart theorem was mentioned to justify the projection theorem. Graduate-level texts give a more comprehensive discussion of the quantum theory of angular momentum, e.g. Cowan (1981), Brink and Satchler (1993) and Sobelman (1996).

## Exercises

(5.1) *Description of the LS-coupling scheme*
Explain what is meant by the central-field approximation and show how it leads to the concept of electron configurations. Explain how perturbations arising from (a) the residual electrostatic interactions, and (b) the magnetic spin–orbit interactions, modify the structure of an isolated multi-electron configuration in the $LS$-coupling limit.

(5.2) *Fine structure in the LS-coupling scheme*
Show from eqn 5.4 that the $J$ levels of the $^3$P term in the 3s4p configuration have a separation given by eqn 5.8 with $\beta_{LS} = \beta_{4\mathrm{p}}/2$ (where $\beta_{4\mathrm{p}}\mathbf{s}\cdot\mathbf{l}$ is the spin–orbit interaction of the 4p-electron).

(5.3) *The LS-coupling scheme and the interval rule in calcium*
Write down the ground configuration of calcium ($Z = 20$). The line at 610 nm in the spectrum of neutral calcium consists of three components at relative positions 0, 106 and 158 (in units of cm$^{-1}$). Identify the terms and levels involved in these transitions.
The spectrum also contains a multiplet of six lines with wavenumbers 5019, 5033, 5055, 5125, 5139 and 5177 (in units of cm$^{-1}$). Identify the terms and levels involved. Draw a diagram of the relevant energy levels and the transitions between them. What further experiment could be carried out to check the assignment of quantum numbers?

(5.4) *The LS-coupling scheme in zinc*
The ground configuration of zinc is 4s$^2$. The seven lowest energy levels of zinc are 0, 32 311, 32 501, 32 890, 46 745, 53 672 and 55 789 (in units of cm$^{-1}$). Sketch an energy-level diagram that shows these levels with appropriate quantum numbers. What evidence do these levels provide that

the $LS$-coupling scheme describes this atom. Show the electric dipole transitions that are allowed between the levels.

(5.5) *The $LS$-coupling scheme*

| 3s3p, Mg | 3s3p, $Fe^{14+}$ |
|---|---|
| 2.1850 | 23.386 |
| 2.1870 | 23.966 |
| 2.1911 | 25.378 |
| 3.5051 | 35.193 |

The table gives the energy levels, in units of $10^6\,\mathrm{m}^{-1}$ measured from the ground state, of the 3s3p configuration in neutral magnesium ($Z = 12$) and the magnesium-like ion $Fe^{14+}$. Suggest, with reasons, further quantum numbers to identify these levels. Calculate the ratio of the spin–orbit interaction energies in the 3s3p configuration of Mg and $Fe^{14+}$, and explain your result. Discuss the occurrence in the solar spectrum of a strong line at 41.726 nm that originates from $Fe^{14+}$. Would you expect a corresponding transition in neutral Mg?

(5.6) *$LS$-coupling for configurations with equivalent electrons*

(a) List the values of the magnetic quantum numbers $m_{l_1}$, $m_{s_1}$, $m_{l_2}$ and $m_{s_2}$ for the two electrons in an $n\mathrm{p}^2$ configuration to show that fifteen degenerate states exist within the central-field approximation. Write down the values of $M_L = m_{l_1} + m_{l_2}$ and $M_S = m_{s_1} + m_{s_2}$ associated with each of these states to show that the only possible terms in the $LS$-coupling scheme are $^1\mathrm{S}$, $^3\mathrm{P}$ and $^1\mathrm{D}$.

(b) The $1\mathrm{s}^2 2\mathrm{s}^2 2\mathrm{p}^2$ configuration of doubly-ionized oxygen has levels at relative positions 0, 113, 307, 20 271 and 43 184 (in units of $\mathrm{cm}^{-1}$) above the ground state, and its spectrum contains weak emission lines at $19\,964\,\mathrm{cm}^{-1}$ and $20\,158\,\mathrm{cm}^{-1}$. Identify the quantum numbers for each of the levels and discuss the extent to which the $LS$-coupling scheme describes this multiplet.

(c) For six d-electrons, in the same sub-shell, write a list of the values of the $m_s$ and $m_l$ for the individual electrons corresponding to $M_S = 2$ and $M_L = 2$. Briefly discuss why this is the maximum value of $M_S$, and why $M_L \leqslant 2$ for this particular value of $M_S$. (Hence from Hund's rules the $^5\mathrm{D}$ term has the lowest energy.) Specify the lowest-energy term for each of the five configurations $n\mathrm{d}$, $n\mathrm{d}^2$, $n\mathrm{d}^3$, $n\mathrm{d}^4$ and $n\mathrm{d}^5$.

(5.7) *Transition from $LS$- to $jj$-coupling*

| $J$ | 3p4s, Si Energy ($10^6\,\mathrm{m}^{-1}$) | $J$ | 3p7s, Si Energy ($10^6\,\mathrm{m}^{-1}$) |
|---|---|---|---|
| 0 | 3.968 | 0 | 6.154 |
| 1 | 3.976 | 1 | 6.160 |
| 2 | 3.996 | 2 | 6.182 |
| 1 | 4.099 | 1 | 6.188 |

The table gives $J$-values and energies (in units of $10^6\,\mathrm{m}^{-1}$ measured from the ground state) of the levels in the 3p4s and 3p7s configurations of silicon. Suggest further quantum numbers to identify the levels.
Why do the two configurations have nearly the same value of $E_{J=2} - E_{J=0}$ but quite different energy separations between the two $J = 1$ states?

(5.8) *Angular-momentum coupling schemes*

| $J$ | 4p5s, germanium Energy ($10^6\,\mathrm{m}^{-1}$) | $J$ | 5p6s, tin Energy ($10^6\,\mathrm{m}^{-1}$) |
|---|---|---|---|
| 0 | 3.75 | 0 | 3.47 |
| 1 | 3.77 | 1 | 3.49 |
| 2 | 3.91 | 2 | 3.86 |
| 1 | 4.00 | 1 | 3.93 |

The table gives the $J$-values and energies (in units of $10^6\,\mathrm{m}^{-1}$ measured from the ground state) of the levels in the configurations 4p5s in Ge and 5p6s in Sn. Data for the 3p4s configuration in Si are given in the previous exercise. How well does the $LS$-coupling scheme describe the energy levels of the $n\mathrm{p}(n+1)\mathrm{s}$ configurations with $n = 3$, 4 and 5? Give a physical reason for the observed trends in the energy levels.
One of the $J = 1$ levels in Ge has a Landé $g$-factor of $g_J = 1.06$. Which level would you expect this to be and why?

(5.9) *Selection rules in the $LS$-coupling scheme*
State the selection rules that determine the configurations, terms and levels that can be connected by an electric dipole transition in the $LS$-coupling approximation. Explain which rules are rigorous, and which depend on the validity of the coupling

scheme. Give a physical justification for three of these rules.

Which of the following are allowed for electric dipole transitions in the *LS*-coupling scheme:

(a) 1s2s $^3S_1$–1s3d $^3D_1$,

(b) 1s2p $^3P_1$–1s3d $^3D_3$,

(c) 2s2p $^3P_1$–2p$^2$ $^3P_1$,

(d) 3p$^2$ $^3P_1$–3p$^2$ $^3P_2$,

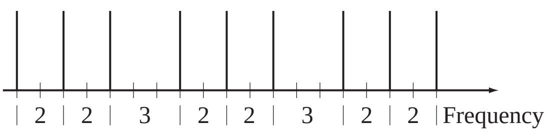
(e) 3p$^6$ $^1S_0$–3p$^5$3d $^1D_2$?

The transition 4d$^9$5s$^2$ $^2D_{5/2}$–4d$^{10}$5p $^2P_{3/2}$ satisfies the selection rules for $L, S$ and $J$ but it appears to involve two electrons jumping at the same time. This arises from configuration mixing—the residual electrostatic interaction may mix configurations.[21] The commutation relations in eqns 5.2 and 5.3 imply that $H_{\mathrm{re}}$ only mixes terms of the same $L, S$ and $J$. Suggest a suitable configuration that gives rise to a $^2P_{3/2}$ level that could mix with the 4d$^{10}$5p configuration to cause this transition.

(5.10) *The anomalous Zeeman effect*
What selection rule governs $\Delta M_J$ in electric dipole transitions? Verify that the $^3S_1$–$^3P_2$ transition leads to the pattern of nine equally-spaced lines shown in Fig. 5.13 when viewed perpendicular to a weak magnetic field. Find the spacing for a magnetic flux density of 1 T.

(5.11) *The anomalous Zeeman effect*
Draw an energy-level diagram for the states of $^3S_1$ and $^3P_1$ levels in a weak magnetic field. Indicate the allowed electric dipole transitions between the Zeeman states. Draw the pattern of lines observed perpendicular to the field on a frequency scale (marked in units of $\mu_B B/h$).

(5.12) *The anomalous Zeeman effect*




The above Zeeman pattern is observed for a spectral line that originates from one of the levels of a $^3P$ term in the spectrum of a two-electron system; the numbers indicate the relative separations of the lines, observed perpendicular to the direction of the applied magnetic field. Identify $L$, $S$ and $J$ for the two levels in the transition.[22]

(5.13) *The anomalous Zeeman effect in alkalis*
Note that atoms with one valence electron are not discussed explicitly in the text.

(a) Give the value of $g_J$ for the one-electron levels $^2S_{1/2}$, $^2P_{1/2}$ and $^2P_{3/2}$.

(b) Show that the Zeeman pattern for the 3s $^2S_{1/2}$–3p $^2P_{3/2}$ transition in sodium has six equally-spaced lines when viewed perpendicular to a weak magnetic field. Find the spacing (in GHz) for a magnetic flux density of 1 T. Sketch the Zeeman pattern observed along the magnetic field.

(c) Sketch the Zeeman pattern observed perpendicular to a weak magnetic field for the 3s $^2S_{1/2}$–3p $^2P_{1/2}$ transition in sodium.

(d) The two fine-structure components of the 3s–3p transition in sodium in parts (b) and (c) have wavelengths of 589.6 nm and 589.0 nm, respectively. What magnetic flux density produces a Zeeman splitting comparable with the fine structure?[23]

(5.14) *The Paschen–Back effect*
In a strong magnetic field **L** and **S** precess independently about the field direction (as shown in Fig. 5.14), so that $J$ and $M_J$ are not good quantum numbers and appropriate eigenstates are $|LM_L SM_S\rangle$. This is called the Paschen–Back effect. In this regime the *LS*-coupling selection rules are $\Delta M_L = 0, \pm 1$ and $\Delta M_S = 0$ (because the electric dipole operator does not act on the spin).[24] Show that the Paschen–Back effect leads to a pattern of three lines with the same spacing as in the normal Zeeman effect (i.e. the same as if we completely ignore spin).[25]

[21] In the discussion of the *LS*-coupling scheme we treated $H_{\mathrm{re}}$ as a perturbation on a configuration and assumed that $E_{\mathrm{re}}$ is small compared to the energy separation between the configurations in the central field. This is rarely true for high-lying configurations of complex atoms.

[22] The relative intensities of the components have not been indicated.

[23] This value is greater than 1 T so the assumption of a weak field in part (b) is valid.

[24] The rules for $J$ and $M_J$ are not relevant in this regime.

[25] The Paschen–Back effect occurs when the valence electrons interact more strongly with the external magnetic field than with the orbital field in $H_{\mathrm{s-o}}$. The *LS*-coupling scheme still describes this system, i.e. $L$ and $S$ are good quantum numbers.